

Auditing Category Quality with Embeddings + Clustering

Learning Objectives

- Explore intra-category structure: do descriptions form meaningful subgroups?
- Practice clustering/visualization on one category to judge label quality.
- Use TSNE and DBSCAN to spot subclusters and potential label noise.
- Sample texts from regions/clusters to interpret what the model is seeing.

Dataset Case Study: My CHI My Future (MCMF) Programs in Chicago

This case study explores real-world data from the My CHI My Future (MCMF) youth programs in Chicago. The dataset was selected for the [MCDC](#) practicum course (Spring 2023, Dr. Lizhen Shi) in partnership with [My CHI My Future](#) and community leader Stender Von Oehsen. The project aimed to address real community needs through data science, with results featured on the [MCDC Past Projects website](#).

Context:

- The MCMF initiative, led by the City of Chicago, connects youth to thousands of out-of-school programs, events, and resources across the city.
- The dataset includes detailed information about program names, descriptions, categories, dates, and locations, providing a rich resource for text analysis and feature engineering.
- Students worked with community partners to clean, analyze, and model this data, supporting efforts to improve youth engagement and access to opportunities.

This notebook demonstrates how to preprocess and analyze text data from a real civic dataset, preparing it for downstream modeling and actionable insights. This lesson was co-developed by **Dr. Lizhen Shi** and **Dr. Arend Kuyper** for the MCDC pathway courses (Summer 2025).

Overview

This notebook zooms into a single category from the cleaned dataset to check how consistent the descriptions are. We cluster within one label, visualize in 2D, and read sample texts to judge quality.

```
import pandas as pd
import numpy as np
```

Load the cleaned dataset

Bring in `datasets/data_cleaned_20250826.csv` from the preprocessing notebook so we work on already-cleaned text fields.

```
df = pd.read_csv('datasets/data_cleaned_20250826.csv', low_memory=False)
df = df[df['Description_Clean'].str.len() > 25]
df['program_name'] = df['Program Name']
df['category_name'] = df['Category Name']
df['description'] = df['Description_Clean']
df = df[['program_name', 'category_name', 'description']].reset_index(drop=True)
df.head()
```

	program_name	category_name	description
0	Homework Help: Virtual Teacher in the Library	Music & Art.	get homework help certified tea
1	Art of Storytelling Learning Circle	Academic Support	makes good story developed pix
2	Learn American Sign Language	Academic Support	want brush american thurgood
3	Learn American Sign Language	Academic Support	want brush american thurgood
4	The Harold Washington Experience: Historians R...	Academic Support	april harold washington history

Check category sizes

Scan counts to decide which category to inspect; start with a large one for clearer structure.

```
df.groupby('category_name').size().sort_values(ascending=False)
```

```
category_name
Sports + Wellness.      121319
Music & Art.            105638
Reading & Writing.      34535
```

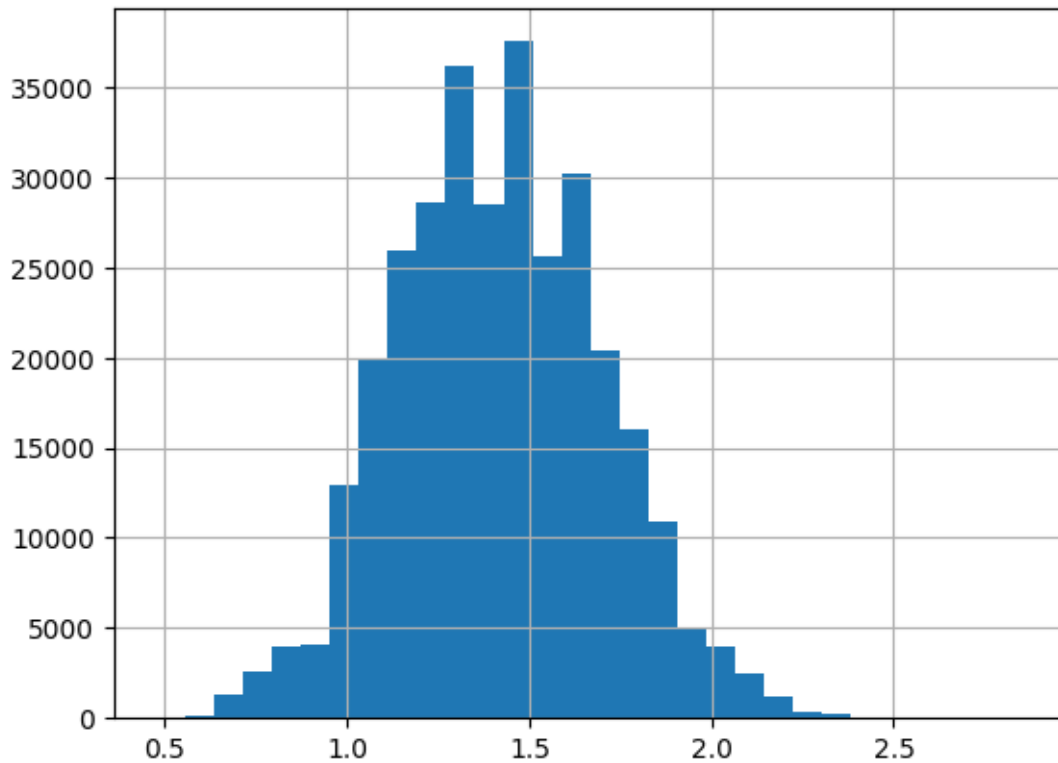
Academic Support	13002
Science	9442
Computers.	9295
Building & Fixing Things	6616
Helping Your Community.	3170
Performance.	2618
Nature.	2569
Managing Money.	2177
Healthcare	2154
Food.	1711
Work + Career	1665
Digital Media.	1556
Social Studies	499
Teaching	167
Customer/Human Service	91
Math	60
Law	48
Transportation	24

dtype: int64

Inspect description length

Length distribution hints at how much text we can rely on and informs any truncation choices.

```
df['description'].map(lambda s: np.log10(len(s.split()))).hist(bins=30)
```



```
df.loc[251513]['description']
```

'everyone gets crafty creative funfilled dabble kinds art material paper art clay projects w

Peek at sample descriptions

Read a handful of truncated descriptions to understand tone and remaining noise.

```
def f(s):
    s=s.split()
    return " ".join(s[:30])
with pd.option_context('display.max_colwidth', None):
    display(df['description'].apply(f).to_frame().sample(10))
```

	description
301949	basketball league playoffs championship games mount greenwood monday
190610	fornbspbook apptit mayfair branches cookbook pick favorite cookbook recipe share groupaccessibility

	description
123747	every summer staff campers look forward reconnecting safe engaging inclusive activitiesthe summer in
207320	universidad popular offers interactive english pleasant environment fill job applications prepare interv
11566	assistive resources talking book booknbspclub discuss happiness projectnbspauthors personal reflection
97647	welcome legler regional maker labnbspdrop shop hours use equipment sewing machines printer laser c
244073	artist maya basics ceramics together well explore handbuilding techniques pinching slab building coili
256118	offers chance explore world dinosaurs making dinomite crafts new friends focus different dinosaur
184833	short stories singing craft toddlers caregiversacompanos leer historias cantar canciones hacer una mar
71772	interested use computer refresher offering oneonone basic computer development resource digital pref

Focus on one category

Filter to a single label (here `Academic Support`) to study within-category consistency. Swap this value to audit other categories.

```
category_name = 'Academic Support'
df = df[df['category_name'] == category_name]
df.shape
```

(13002, 3)

```
description_cutoff = 30
```

Truncate descriptions to the first 30 tokens to keep embeddings fast and comparable.

Create embeddings

Generate dense vectors for each filtered description using `fastembed.TextEmbedding`.

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.decomposition import LatentDirichletAllocation
from sklearn.cluster import KMeans
import numpy as np
from fastembed import TextEmbedding

embedding_model = TextEmbedding()

embedding_matrix = list(embedding_model.embed(df['description']).map(lambda s: " ".join(s.split(" "))))
```

```
embedding_matrix = np.array(embedding_matrix)
embedding_matrix.shape
```

(13002, 384)

Choose k with an elbow plot

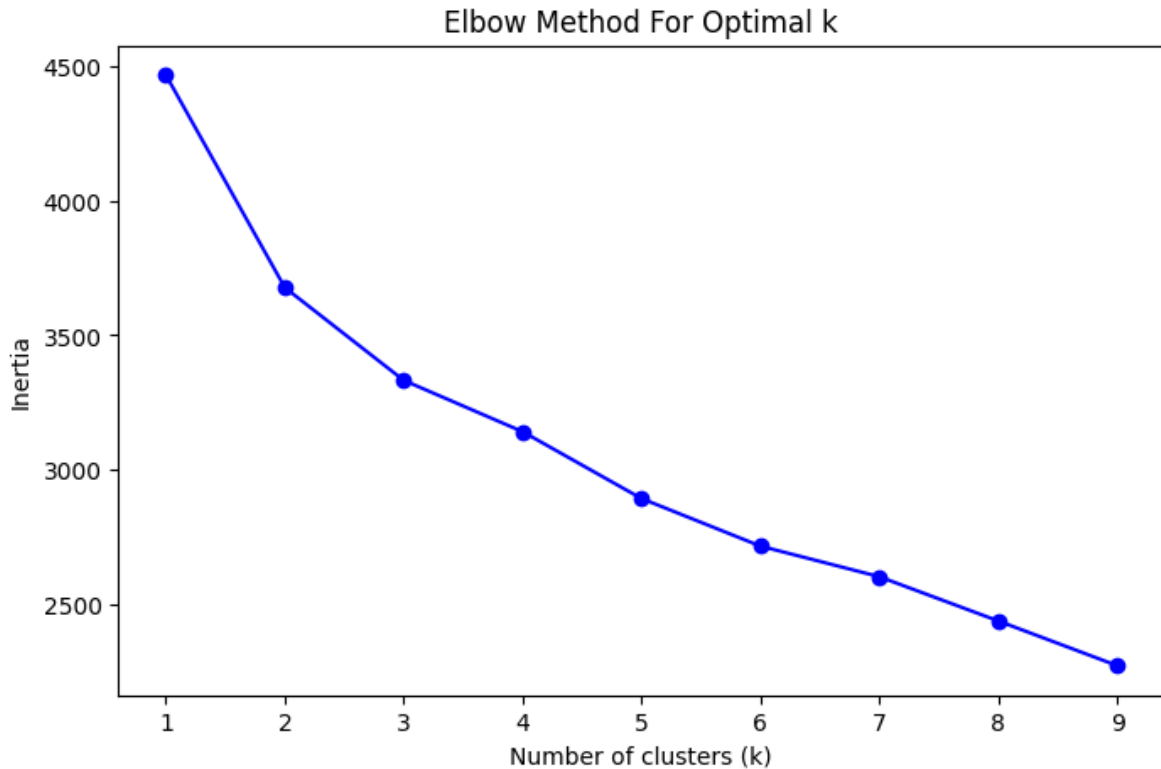
Use inertia across k values to pick a reasonable cluster count before visualizing.

```
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Elbow method
inertias = []
K = range(1, 10) # test k values from 2 to 14

for k in K:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(embedding_matrix)
    inertias.append(km.inertia_)

# Plot
plt.figure(figsize=(8, 5))
plt.plot(K, inertias, 'bo-')
plt.xlabel('Number of clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method For Optimal k')
plt.show()
```



Cluster within the category

Run KMeans at the chosen k and reduce dimensions with SVD for a quick 2D view.

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans
from sklearn.decomposition import TruncatedSVD
import matplotlib.pyplot as plt

# Suppose you already have:
# tfidf_matrix = vectorizer.fit_transform(docs)

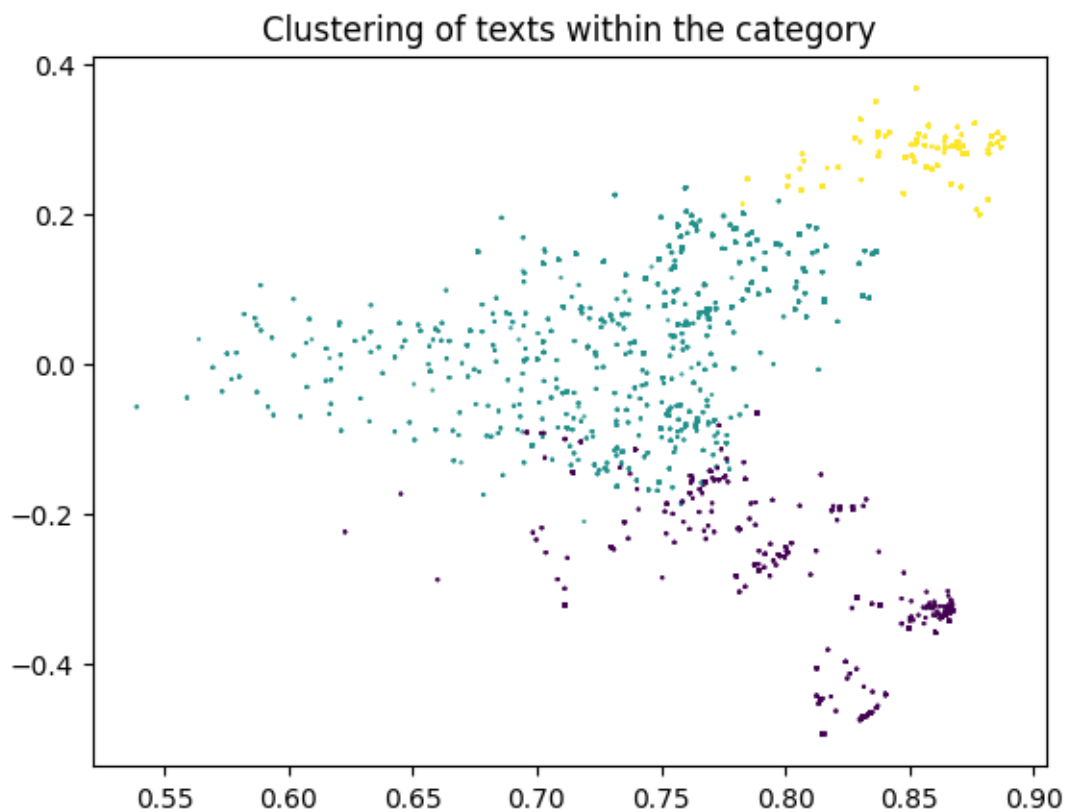
# 1. Look at basic info
print("Shape of TF-IDF matrix:", embedding_matrix.shape)  # (num_docs, num_terms)

# 2. Cluster the docs
n_clusters = 3 # you can adjust or use silhouette score to find a good k
km = KMeans(n_clusters=n_clusters, random_state=42)
clusters = km.fit_predict(embedding_matrix)
```

```
# 3. Dimensionality reduction for visualization
svd = TruncatedSVD(n_components=2, random_state=42)
reduced = svd.fit_transform(embedding_matrix)

plt.scatter(reduced[:, 0], reduced[:, 1], c=clusters, cmap="viridis", s=0.3)
plt.title("Clustering of texts within the category")
plt.show()
```

Shape of TF-IDF matrix: (13002, 384)



Visualize with TSNE

Project embeddings with TSNE to check nonlinear structure and cluster separation.

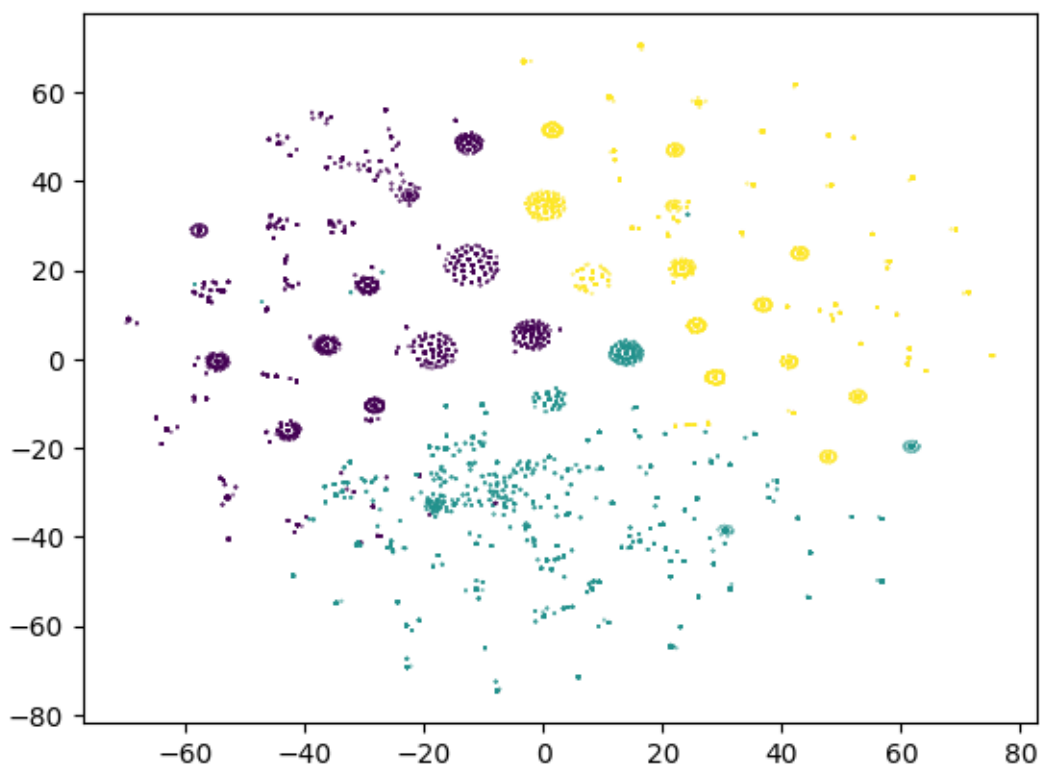
```
from openTSNE import TSNE

embedding = TSNE(n_jobs=-1).fit(embedding_matrix)
embedding.shape
```

(13002, 2)

Color TSNE by KMeans labels to see how clusters map in 2D.

```
plt.scatter(embedding[:, 0], embedding[:, 1], c=clusters, cmap="viridis", s=0.1)
```



DBSCAN on TSNE space

Run density-based clustering on the TSNE embedding to spot fine-grained groupings or noise points.

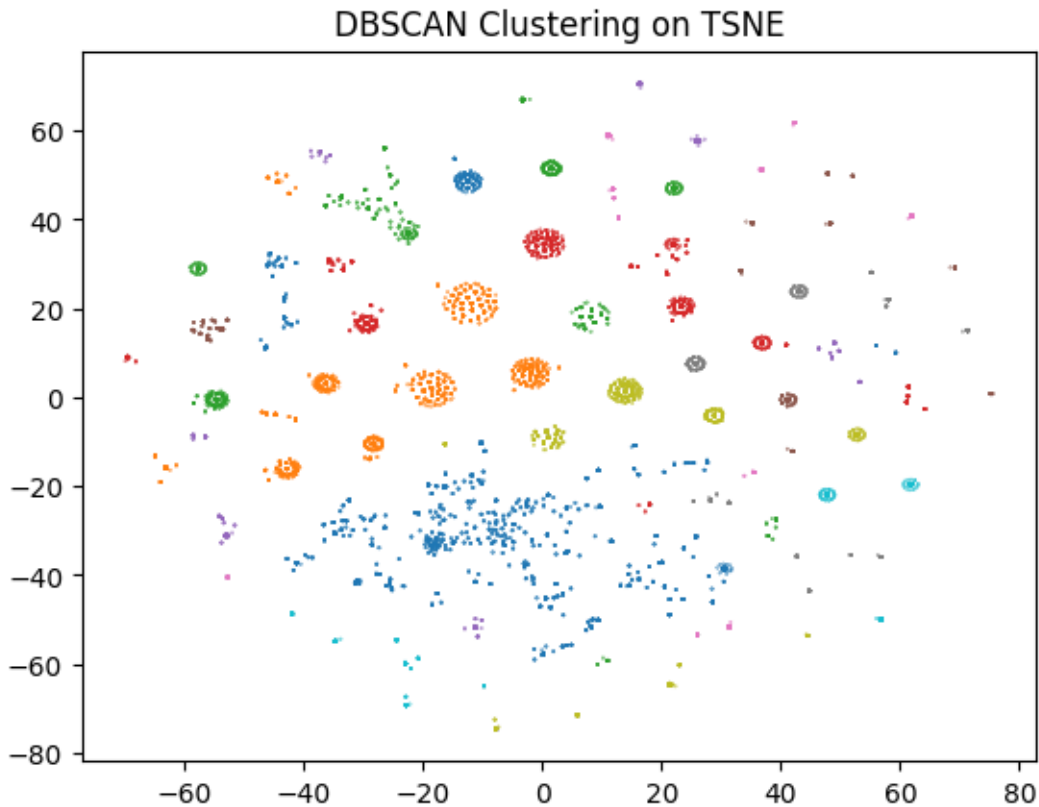
```

from sklearn.cluster import DBSCAN

dbscan = DBSCAN(eps=5, min_samples=10)
tsne_clusters = dbscan.fit_predict(embedding)

plt.scatter(embedding[:, 0], embedding[:, 1], c=tsne_clusters, cmap="tab10", s=0.1)
plt.title("DBSCAN Clustering on TSNE")
plt.show()

```



```
tsne_clusters
```

```
array([0, 0, 0, ..., 0, 0, 0], shape=(13002,))
```

Attach cluster labels back to the data

Store KMeans clusters, TSNE coordinates, and DBSCAN clusters for inspection.

```

df['cluster'] = clusters
df['dim_0'] = reduced[:, 0]
df['dim_1'] = reduced[:, 1]
df['tsne_0'] = embedding[:, 0]
df['tsne_1'] = embedding[:, 1]
df['tsne_clusters'] = tsne_clusters
df['cluster'].value_counts()

```

```

cluster
0      4504
1      4495
2       4003
Name: count, dtype: int64

```

Check how many points fall into each DBSCAN cluster; lots of -1 values can indicate noise or overly strict eps/min_samples.

```

df['tsne_clusters'].value_counts()

```

```

tsne_clusters
0      1877
17       557
10       486
2        449
18       380
...
37        27
55        26
87        24
57        24
74        21
Name: count, Length: 89, dtype: int64

```

Read sample texts by region

Sample descriptions from different SVD dimensions to interpret what the components capture.

```
with pd.option_context('display.max_colwidth', None):
    display(df[(df['dim_1'] > 0.1)].sample(5))
    display(df[(df['dim_1'] < 0)].sample(5))
```

	program_name	category_name
12040	English Classes for Intermediate Levels/Clases de Inglés de Nivel Intermedio	Academic Support
29898	ESL - English Language Conversation Practice	Academic Support
300357	French Conversation Tuesdays	Academic Support
73938	English as a Second Language (ESL) Class	Academic Support
147875	English as Second Language (ESL) Workshop	Academic Support

	program_name	category_name	description
277161	Kiddie College at Euclid	Academic Support	recreational based preschoollike provide
5123	Park Kids - Winter at Lake Shore	Academic Support	safe grammar age providing variety sp
100893	Kiddie College at Revere	Academic Support	visit park district website view availab
253284	Moms, Pops & Tots Interaction at Hale	Academic Support	visit park district website view availab
176899	Tiny Talkers - Spanish at Eugene Field	Academic Support	spanish intended ages old interactive i

Read sample texts by DBSCAN cluster

Pull example descriptions per TSNE/DBSCAN cluster to label them qualitatively and spot misfits.

```
with pd.option_context('display.max_colwidth', None):
    display(df[df['tsne_clusters']==2].sample(5))
    display(df[df['tsne_clusters']==10].sample(5))
    display(df[df['tsne_clusters']==18].sample(5))
```

	program_name	category
148487	Clases de Inglés - English as a Second Language Classes	Academic
146660	English as Second Language (ESL) Workshop	Academic
141719	Afternoon Spanish Conversation Group: Grupo de conversación en español por la tarde	Academic
139787	Spanish Conversation Group	Academic
298496	Spanish Conversation Circle	Academic

	program_name	category_name	description
312875	Kiddie College at Mann	Academic Support	recreational based preschoollike provi
149942	Kiddie College at Independence	Academic Support	recreational based preschoollike provi
67300	Kiddie College at Haas	Academic Support	recreational based preschoollike provi
233126	Kiddie College at Shabbona	Academic Support	recreational based preschoollike provi
4299	Kiddie College - Spring at Maggie Daley	Academic Support	recreational based preschoollike provi

	program_name	category_name	description
223624	Play School at Merrimac	Academic Support	play provides opportunity socialization interaction p
235144	Play School at West Lawn	Academic Support	play provides opportunity socialization interaction p
252609	Play School at Norwood	Academic Support	play provides opportunity socialization interaction p
59965	Play School at Sheil	Academic Support	play provides opportunity socialization interaction p
35367	Play School at Addams	Academic Support	play provides opportunity socialization interaction p

Wrap-up

You inspected substructure within a single category using embeddings, KMeans, TSNE, and DBSCAN, then read sample texts to interpret clusters. Repeat this for other categories to flag noisy labels, merge overlapping subtopics, or tune clustering parameters (k, eps, min_samples, token cutoff).